



▶ RoboSim

Project team : Freya, Cassie, Raymond, Mincheol, Ziyang

► OUR PROJECT

- Marius Jurt: Masters Student at Bristol, Research Engineer at Toshiba.
- Self-moving, self-stacking robotic boxes.
- RoboSim is a simulation platform for algorithm developers.
- Ranks all algorithms on leaderboard based on time/moves/energy used.



► USER STORIES



WAREHOUSE MANAGER

I want to optimise storage and retrieval of goods, so that I can increase efficiency to reduce the overheads of inventory management.



ALGORITHM DESIGNER

I want to develop and benchmark routing algorithms with an intuitive API, so that I can streamline testing and optimisation of robot navigation in 3D space.

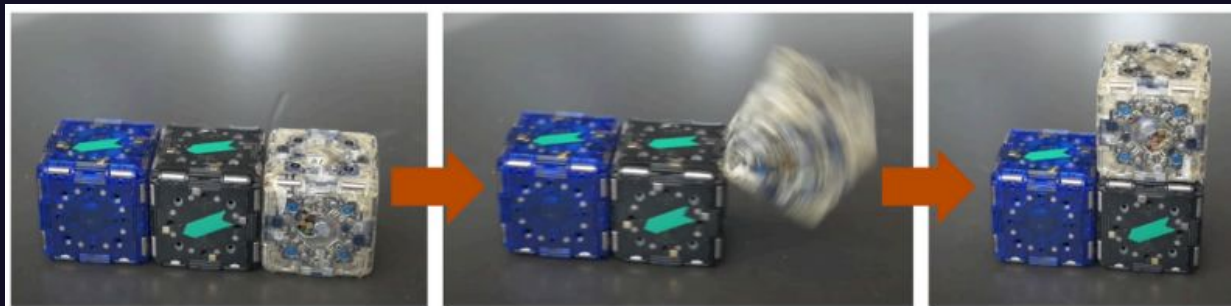


FUTURE CONTRIBUTOR

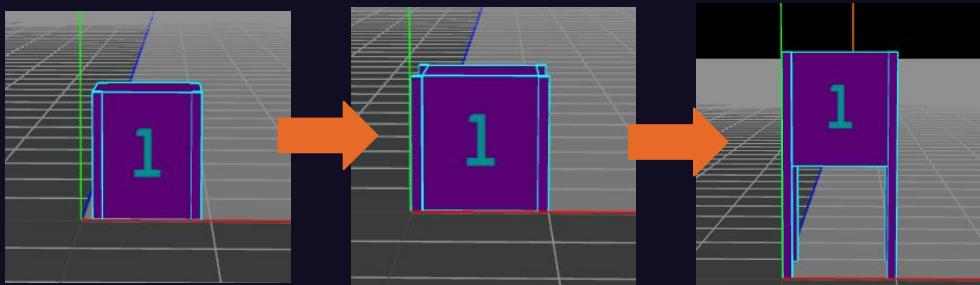
I want support with understanding the existing codebase and project.

► BOX TYPES

TYPE 1



TYPE 2





► **DEMO!!**

Tech Stack

Web Client:



 TypeScript



Database:



Server:

 TypeScript



Algorithm Client:



► Improvements since last term

- Rotating roles
- Agile development
 - Rotating manager
 - Sprints
- Giving more in depth feedback and code reviews on our PRs

Week	Project Manager
Week 13 - Testing Day	Cassie Lambert
Testing Day - Beta Release	Raymond Kellner
Beta Release - Viva	Ziyang Wang
Viva - Week 21	Mincheol Shin
Week 21 - Week 23	Freya Donnelly

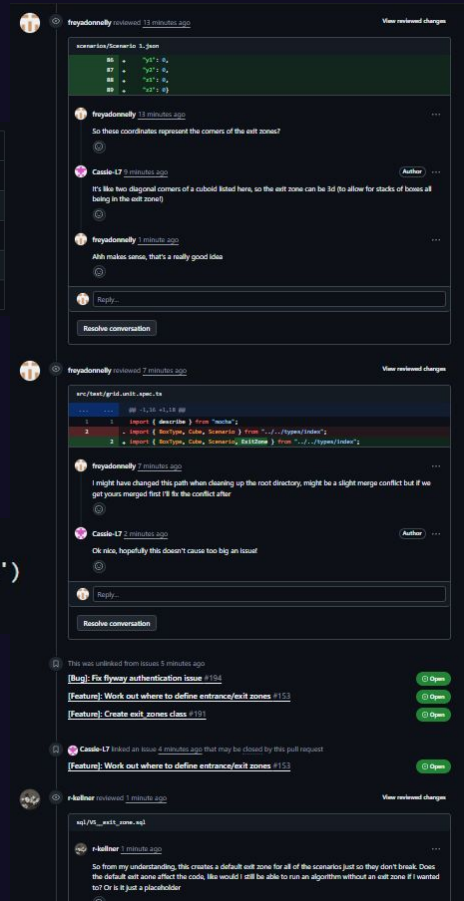
Reflection from last sprints 'improvements':

What could have gone better?

- Bugs with root directory - but cant control that :')
- Updating more frequently rather than big commits



	Author	Label	Projects	Milestones	Reviews	Assignee	Sort
Clean up root directory					2		6
#193 opened yesterday by freyadonnelly 4 of 12 tasks Beta Release							
Exit zones					2		13
#190 opened 2 days ago by Cassie-L7 5 of 12 tasks Beta Release							
Clearer Algorithm Client documentation + Raymond ai-docs					2		5
#187 opened 5 days ago by r-kellner 5 of 12 tasks Beta Release							





**▶ CHANGES WE'VE
MADE!**

► Database

Before



After



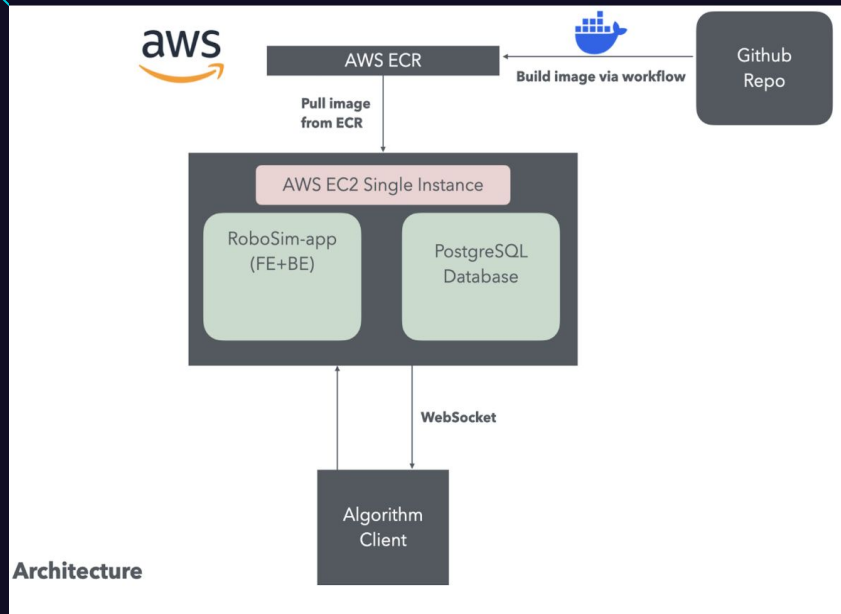
Challenges:

- Slight syntax differences
- Changes needed in so many files!
- SQLite was a file based database, but Postgres needed a container on the server for deployment

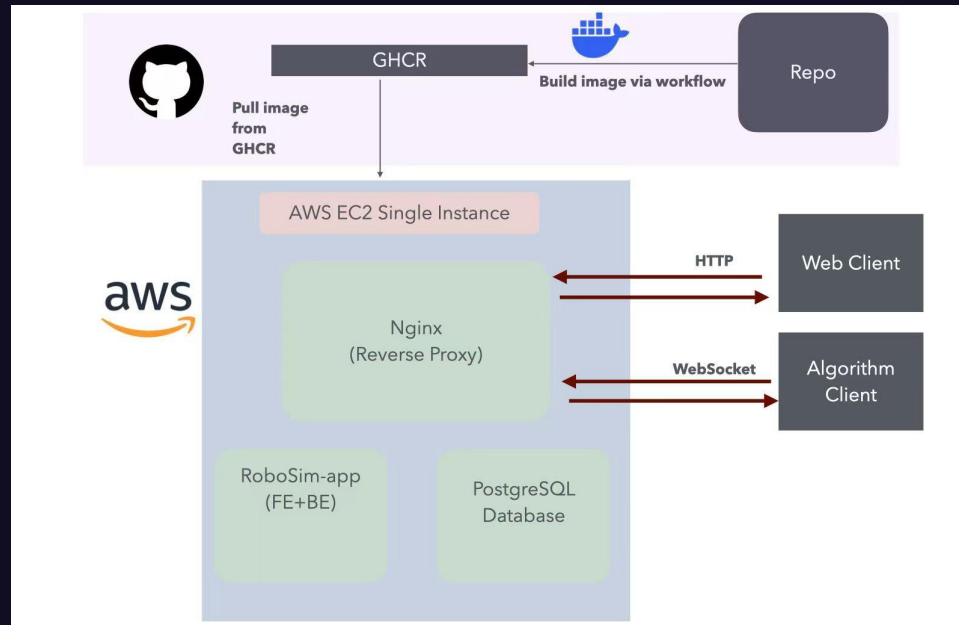
► Deployment CI/CD

Our server web: <http://13.49.70.251/>

Before:



After:

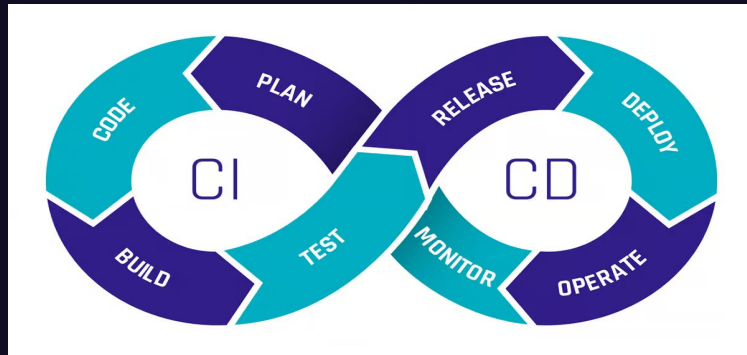


► CODING TEST

We analysis our tests coverage and find that

- Unit test : 121 tests
- Coverage: (compared the final stage of last year 91% in total)

Now we have 93.49%



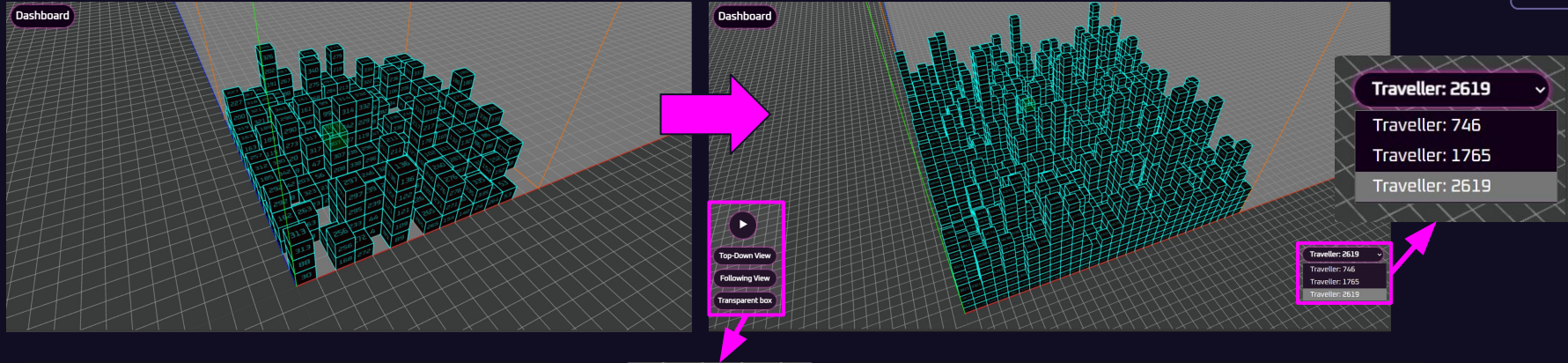
Issue we have met this year:

- Server changes caused lots of logic used before should be optimised or even rewritten.
- Version issues with the technology stack can also lead to changes in test conditions.
- Change project structure will cause lots of path issue in tests. And all the tests should be checked path correctness.
-

► Interactive control features

Before

After

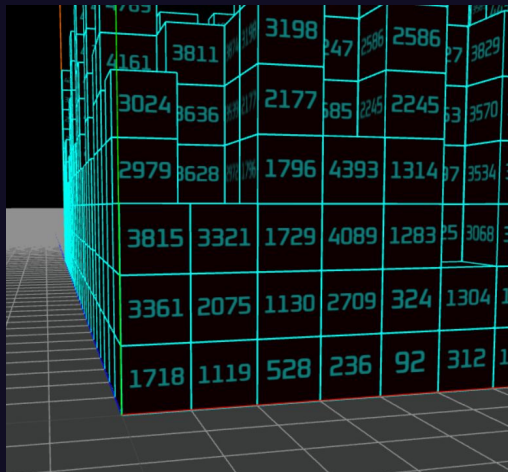


Previously, users had to **manually control** the camera to locate and follow the 'traveller' box. This manual process became **highly inefficient and difficult** as the warehouse scale and the number of boxes increased, making it nearly impossible to monitor the simulation effectively.

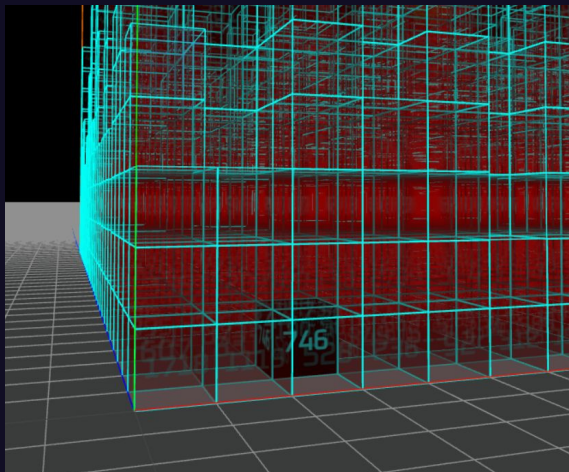
We implemented five core controls to enhance monitoring:
Start/Stop for timing,
Top-down and Following views for perspective,
Transparency for visibility,
and **Dropdowns** for precise traveller box selection.

► Transparent / perspective features

Before

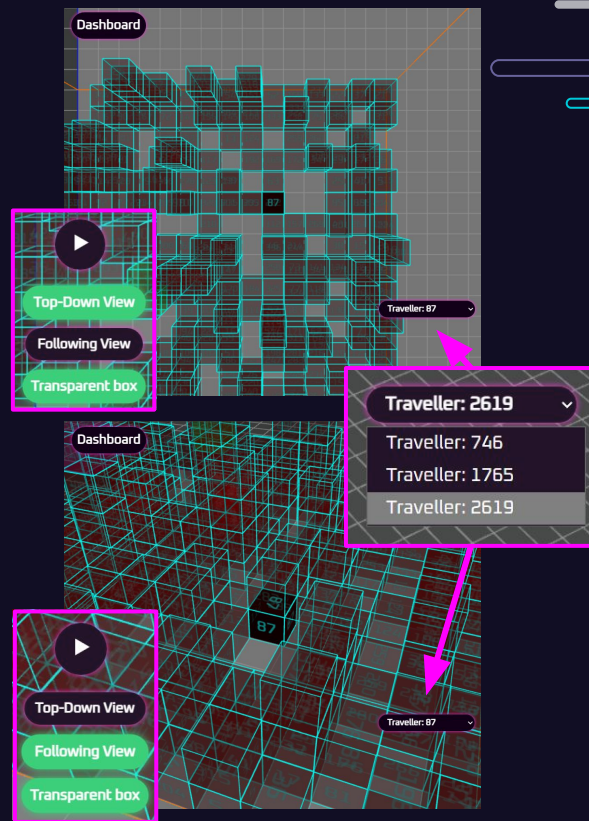


After



Set `depthWrite` to **false** for non-traveller boxes so they don't overwrite the depth buffer. By setting only the traveller box's `depthWrite` to **true**, it is the only one recorded in the buffer, making it visible through other boxes via `depthTest`.

By collect traveller ID from replay record or by request to server, it lists travellers' ID to select on dropdown and target it on one of the perspective mode. Then it directly moves into targeted traveller box when selected.

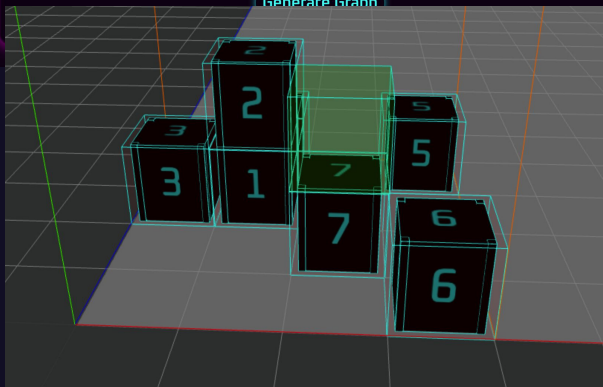


After

Algorithm Client Summary

Scenario	State	Energy Used	View
Scenario 11 - Type 2	Completed in 5.0s	5	View replay
Scenario 13 - Type 2	Pending	0	Follow run
Scenario 12 - Type 2	Completed in 4.0s	3	View replay

Generate Graph



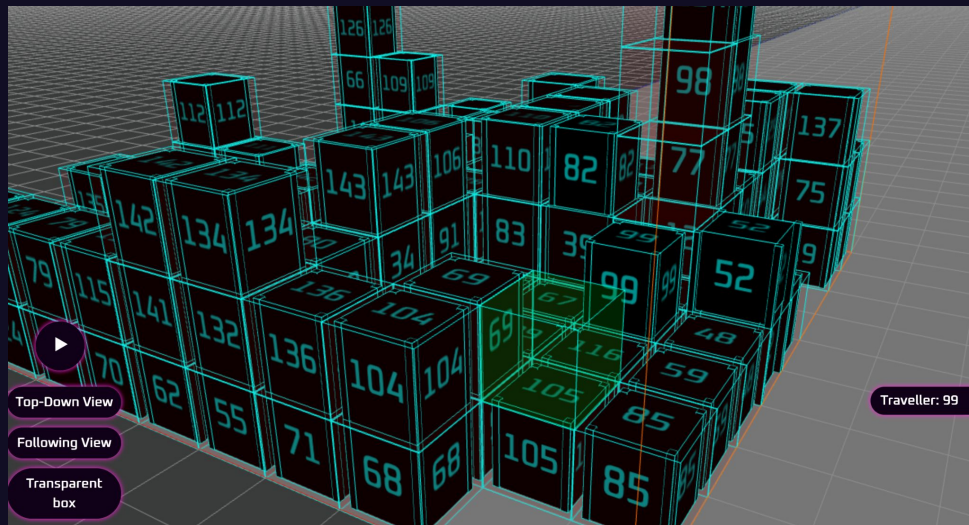
Now able to solve much more complicated scenarios

Algorithm Client Summary

Name: Type 2 Author: Raymond Kellner State: Completed

Scenario	State	Energy Used	View
Scenario 11 - Type 2	Completed in 6.0s	5	View replay
Scenario 13 - Type 2	Completed in 14.1s	13	View replay
Scenario 12 - Type 2	Completed in 4.0s	3	View replay
Scenario 14 - Type 2	Completed in 17.1s	5	View replay
Scenario 15 - Type 2	Completed in 44.3s	61	View replay

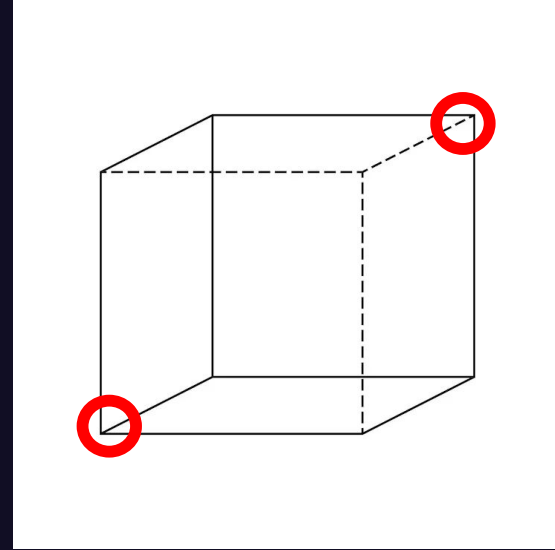
Generate Graph



Traveller: 99

► Exit/entrance zones

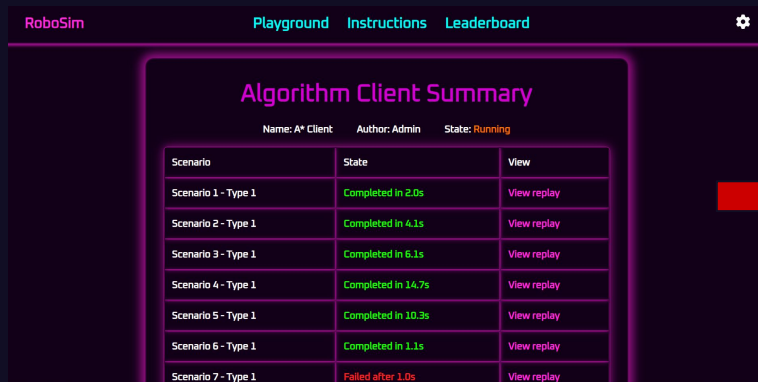
- Important part of warehouses previously missing
- Defined as 2 corners of a cuboid
- WIP, need frontend
- Aim to create algorithms incorporating these



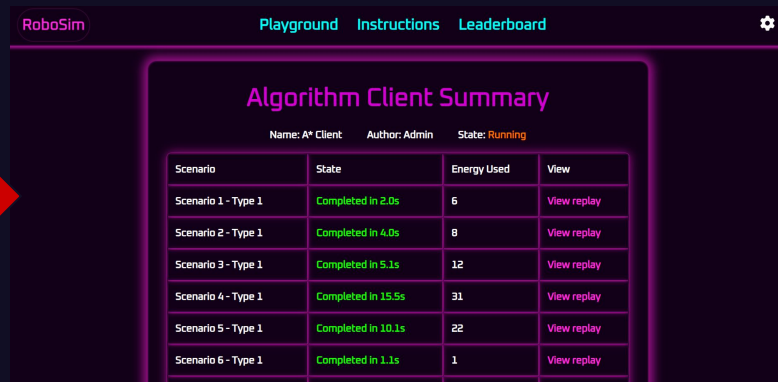
► Energy usage and benchmarking

Before

After



Algorithm Client Summary		
Name: A* Client Author: Admin State: Running		
Scenario	State	View
Scenario 1 - Type 1	Completed in 2.0s	View replay
Scenario 2 - Type 1	Completed in 4.1s	View replay
Scenario 3 - Type 1	Completed in 6.1s	View replay
Scenario 4 - Type 1	Completed in 14.7s	View replay
Scenario 5 - Type 1	Completed in 10.3s	View replay
Scenario 6 - Type 1	Completed in 1.1s	View replay
Scenario 7 - Type 1	Failed after 1.0s	View replay



Algorithm Client Summary			
Name: A* Client Author: Admin State: Running			
Scenario	State	Energy Used	View
Scenario 1 - Type 1	Completed in 2.0s	6	View replay
Scenario 2 - Type 1	Completed in 4.0s	8	View replay
Scenario 3 - Type 1	Completed in 5.1s	12	View replay
Scenario 4 - Type 1	Completed in 15.5s	31	View replay
Scenario 5 - Type 1	Completed in 10.1s	22	View replay
Scenario 6 - Type 1	Completed in 1.1s	1	View replay

Working from feedback given in the MVP, we have added an energy usage tracker to our system to track how much energy an algorithm uses when running on scenarios.

Testing day

Feedback	Resolution	Justification	Area	Ease	Priority
"Could add directions (north/south or door/entrance)"	Add entrance/exit zones to the warehouse simulation for boxes to be moving towards.	Makes the playground a more clear and real representation of a warehouse	Algorithms	Hard	High
"Maybe add some shelves with boxes to mimic a warehouse"	Makes it a clearer representation of a warehouse.	Not that type of representation of a warehouse as boxes are designed to stack on top of each other.	Frontend	Medium	Low
"Depending on use cases, maybe a <u>first person</u> view or a more warehouse like background would indicate more context to the user"	Makes it a <u>more clear</u> representation of a warehouse.	Gives the user context that the playground area is where they simulate the warehouse. <u>However</u> system is currently <u>first person</u> view.	Frontend	Medium	Low
Add "an optional tutorial" when simulating the boxes	Useful to guide the system through how to use the platform.	Means users know how to navigate and do stuff when opening the platform for the first time.	Frontend	Hard	High

Include "clearer instructions on how to move boxes"	Allows the user to easily navigate the system	Will improve the usability of the system	Frontend	Medium	High
Adding functionality of "being able to place blocks not only in the origin square but also any other square, maybe and insert mode for adding boxes and a moving mode for moving boxes"	Would mean when adding boxes they're not just stacked on top of one another which leads a new user to confusion about how to unstack and move etc	Will really improve the ease of use.	Frontend	Medium	Medium
"Ability to replay simulation without reloading the page"	Means replay will function correctly	Definitely a bug to fix, improves usability	Backend	Medium	Medium
"Maybe different shaped boxes so not all cubes"	Additional feature that could be simulating a warehouse that isn't using the MIT box types that Robosim models	Could be inefficient if we're modelling shapes that don't stack optimally, defeating the point of optimally moving the boxes to target locations	Backend/Frontend	Hard	Low
Being able to drag, compress and expand the instructions blob	Improves usability	Currently blocks the playground area meaning you <u>can't</u> work through <u>it</u> side	Frontend	Medium	Medium

Feedback analysed and broken down into a table so we know what we should focus on

"Have a floating list of algorithms when simulating the boxes"	Improves ease of use	Means we don't have to navigate back to preview other algorithms or replays	Frontend	Medium	Low
"Having a FAQs question symbol across all pages"	For bugs/how to use the page	Will improve usability	Frontend	Easy	Low
"Including breakpoints in runs of algorithms"	For developers to be able to debug when an algorithm gets stuck and be able to see the state it stops at	Makes it easier for algorithm designers to simulate and improve their algorithms	Backend/Frontend	Medium	Low
"State-saving on the website"	Adding a "back button" and not resetting/refreshing the page when we go back	When in the replay page - when we click back, it should persist with the last tab we were at	Backend/Frontend	Medium	Medium
" <u>clearer</u> instructions"	Will fix the lack of clarity when learning to use the system	Something we need to improve	Frontend	Easy	High
"Difficult to understand for a new user what's going on. Would be nice to see some kind of feedback that a simulation	Adding some kind of feedback on the screen that the simulation has finished	Would be an easy addition to the system to improve usability	Frontend	Easy	Medium
finished running because they were very short and abruptly stopped."					
"I believe something to work on is how visible the boxes are when they are hidden behind"	Not sure how to fix yet but good thing to work on	Fixes error of box lines being in front of target box in transparent mode	Frontend	Medium	Medium
"Maybe more description is needed for how and why to use this very cool website."	Add to main page context behind Robosim	Will highlight how impressive Robosim is and the motivation behind it for new users	Frontend	Easy	Low

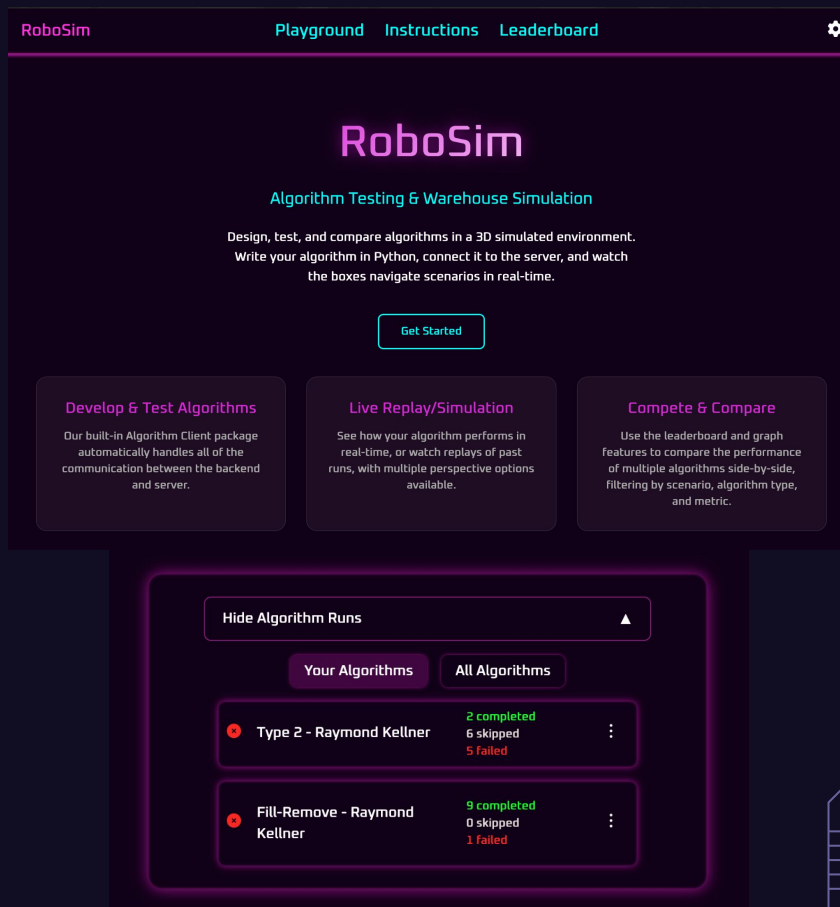
► Dashboard

Before



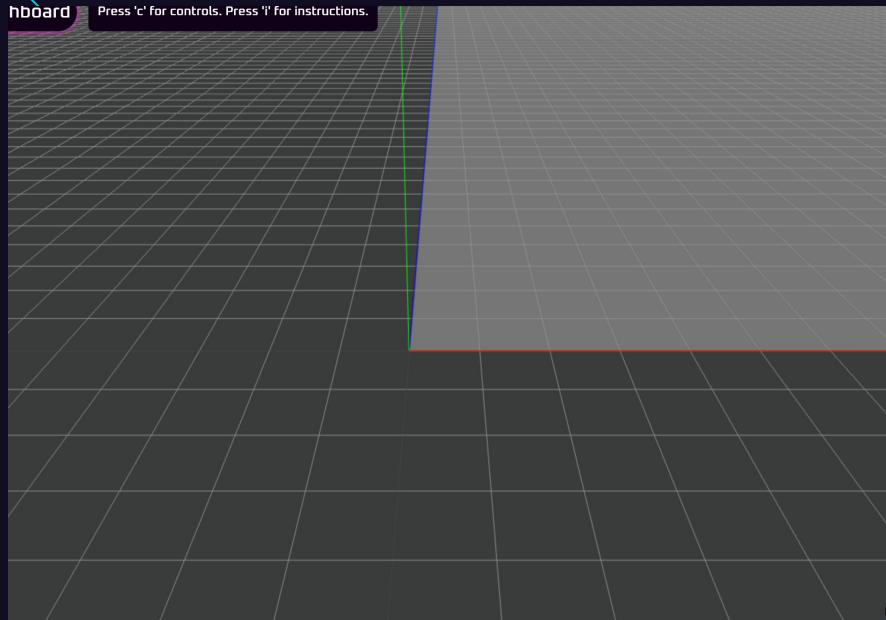
- Transformed the dashboard into a user-friendly landing page
- Hid the algorithm runs behind a drop down menu to visually de-clutter

After



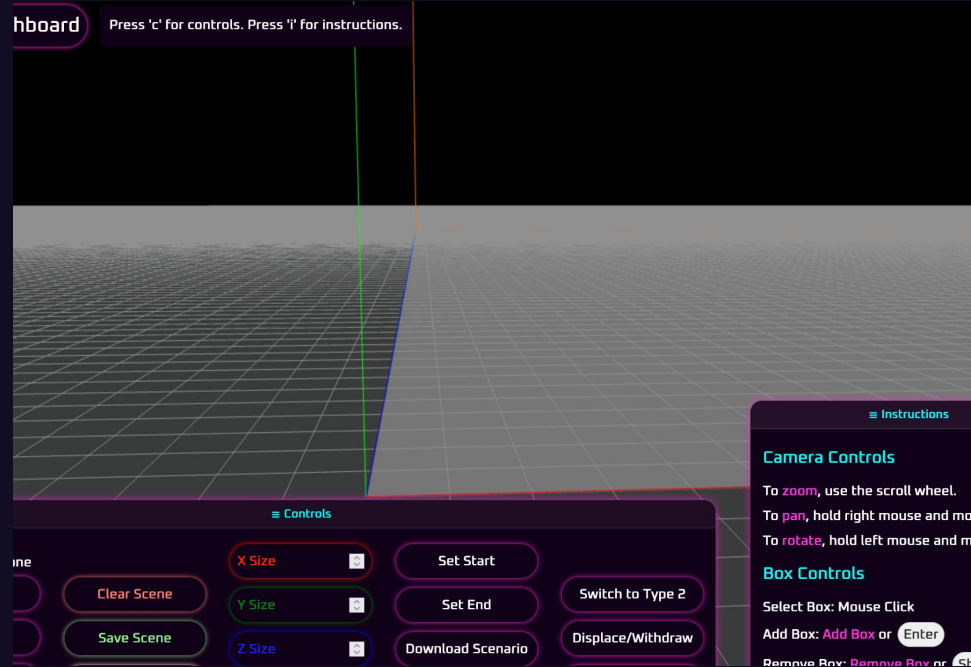
► Playground

Before



After

- Auto loaded controls and instructions
- Added drag handles to both boxes



► Leaderboard

Before

Leaderboard

This shows different sets of benchmarking across different scenarios.

Scenario: ✓ All Scenarios Generate Graph

Sort: Scenarios Completed, Average Time ▼

Name	Author	Scenarios Completed	Average Moves	Average Time Taken	Average Energy Used
Fill-Remove	Raymond Kellner	9/20			

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After

Leaderboard

This shows different sets of benchmarking across different scenarios.

Scenario: All Scenarios Generate Graph

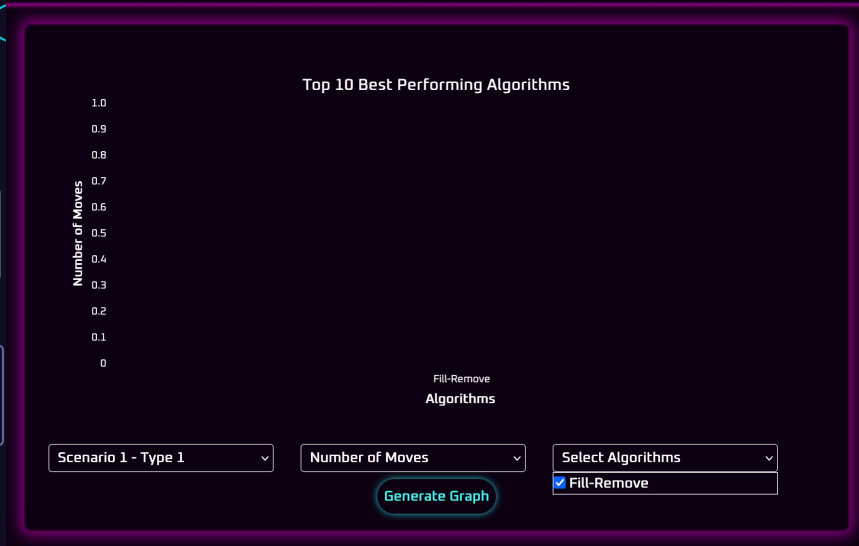
Sort: Scenarios Completed, Average Time ▼

Name	Author	Scenarios Completed	Average Moves	Average Time Taken	Average Energy Used
Fill-Remove	Raymond Kellner	9/20	40.6	26.4s	40.6
Type 2	Raymond Kellner	2/20	8.0	5.5s	8.0

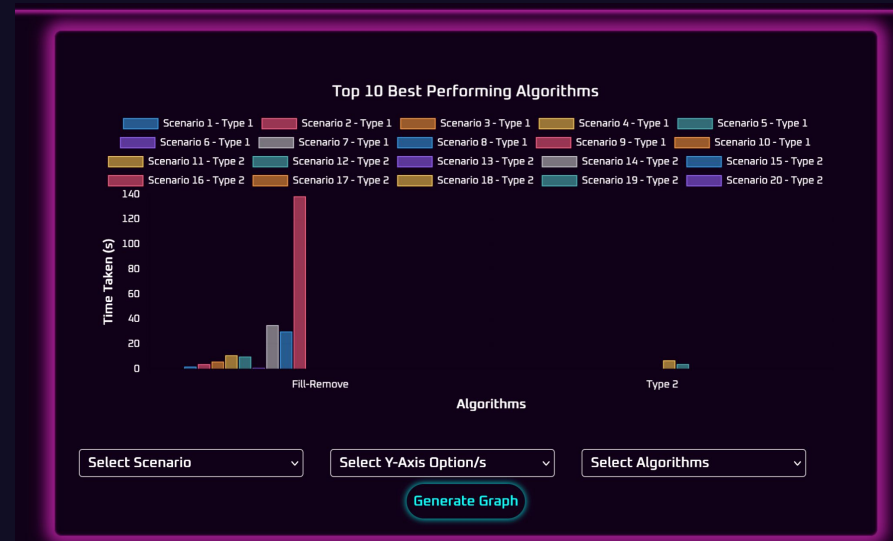
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► Leaderboard

Before

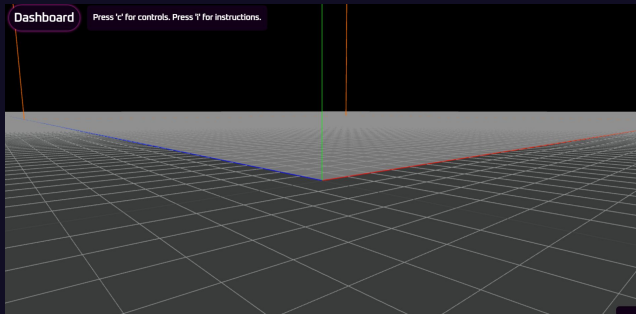
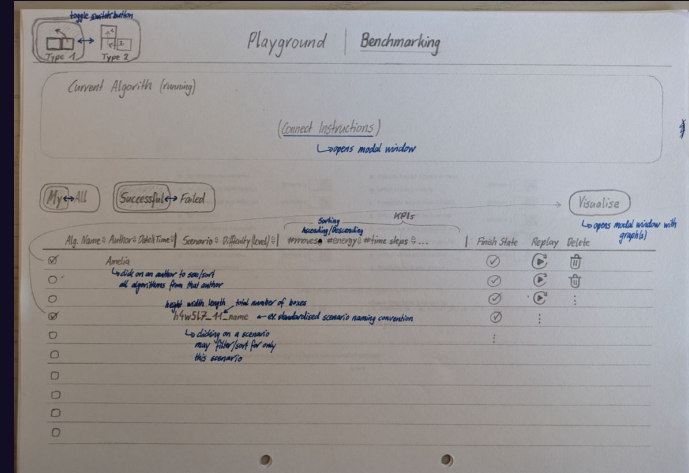


After



► Client feedback

- Leaderboard improvements
- Fixing the playground view range.
- Make it so that we are able to save scenarios.
- Carry on improving exit zones.
- Make a roadmap of our plan.



► Plan for next phase

- Exit zones - frontend, algorithms
- Benchmarking
- Possibly run scenarios and algorithms through the website
- Make replay restart after an algorithm ends
- Changing meeting times to work off weekly feedback